

Competitive Dynamic Pricing · Nash Equilibria

WEEK 1 OF 8

Multi-Agent RL · Oligopoly Simulation · Collusion Detection

PHASE 1 — FOUNDATIONS

Week 1: Game Theory Foundations & Problem Framing — 100% theory this week — no code. Students who skip the pen-and-paper Nash derivation always struggle to interpret their Week 6 results.

Priority key:

**MUST WATCH /
READ**

RECOMMENDED

OPTIONAL

■ = estimated time | pp. = page range

■ YouTube & Interactive Videos

■ Game Theory: The Science of Decision-Making

TED-Ed

youtube.com/watch?v=MHS-htjGgSY

Perfect 5-min opener. Dominant strategies + Prisoner's Dilemma with clear animation. Show at the very first session.

MUST WATCH

■ 5 min

■ Yale Open Course — Game Theory, Lecture 1: Introduction

YaleCourses / Ben Polak

youtube.com/watch?v=nM3rTU927io

Gold-standard intro. Dominant strategies → iterated elimination → Nash Equilibrium. Watch at 1.25×

MUST WATCH

■ 75 min

■ Yale Open Course — Game Theory, Lecture 3: Nash Equilibrium

YaleCourses / Ben Polak

youtube.com/watch?v=lrCqCCYMOiw

Best-response functions and formal NE definition. Derive Bertrand NE on paper immediately after watching.

MUST WATCH

■ 75 min

■ Bertrand Competition Model Explained

EconplusDal

youtube.com/watch?v=sN9GmYIJsaE

Crisp explanation of the Bertrand duopoly — the exact economic model this project is built on. Covers why prices fall to marginal cost.

MUST WATCH

■ 12 min

■ Stackelberg Competition — Game Theory

Economics Explained Properly

youtube.com/watch?v=sMbBIEJNM2c

Covers the Stackelberg leader-follower variant — useful context before designing the market environment in Week 2.

RECOMMENDED

■ 14 min

■ The Evolution of Trust (Interactive Game)

Nicky Case

ncase.me/trust

Interactive browser game. The single best intuition-builder for cooperation/defection and Tit-for-Tat dynamics. Free.

RECOMMENDED

■ 30 min

■ Papers, Books & Blogs

■ A Course in Game Theory — Ch. 2 & 3

Osborne & Rubinstein

theory.economics.utoronto.ca (free PDF)

Formal NE definition and existence theorem. Derive the Bertrand equilibrium by hand after finishing Ch. 3.

MUST READ

■ ~90 min

pp. 11–46

■ AI, Algorithmic Pricing and Collusion

Calvano, Calzolari, Denicolò, Pastorello — QJE 2020

doi.org/10.1093/qje/qjz036

The paper this project is based on. Read intro + Section 2 this week. Return to Sections 3–5 in Weeks 4 and 6.

MUST READ

■ ~45 min

pp. 1–12 (intro + Sec. 2)

— Nash Equilibrium — Intuitive Explainer with Examples

Towards Data Science

towardsdatascience.com → search 'Nash Equilibrium'

15-min read. Covers Prisoner's Dilemma, Bertrand, and Cournot with worked Python examples. Good warm-up before the formal Yale lectures.

RECOMMENDED

■ ~15 min

Web article

■ **An Introduction to Game Theory — Ch. 1 & 2**

Martin Osborne
economics.utoronto.ca/osborne/igt

Gentler on-ramp if Osborne & Rubinstein feels too dense. Less formal, more intuition-first — good for students with weaker economics backgrounds.

OPTIONAL
■ ~60 min
pp. 1–30

■ **Common Pitfalls to Watch — Week 1**

- Confusing Nash Equilibrium with Pareto Optimality — address this early; they are fundamentally different concepts.
- Scope creep: 'let's also model auctions' — keep strict focus on Bertrand/Stackelberg duopoly only.

■ Videos (must)	■ Videos (rec.)	■ Reading (must)	■ Total (must only)	■ Total (all items)
167 min	44 min	135 min	~5 hr	~8 hr